

# Drilling Mechanics - APWD

## Time Based

Real Time, Recorded Mode, Run 9

# Schlumberger

Company: Equinor

Well: 31/5-7

Field: Eos

Rig Name: West Hercules

Country: Norway

Latitude: 60° 34' 35.14" N

Longitude: 3° 26' 36.13" E

Block: 31/5

UWID:

Rig Name:

Rig Type:

West Hercules

Semi-Submersible

FL: Norwegian North Sea

FL1: Section: 8 1/2 in.

FL2: Job no.: 19SCA0085

Log Measured From: - Drill Floor: 31.00 m  
Permanent Datum: - Mean Sea Level

Ground Level: 307.00 m

Acquisition Dates: 29-Dec-2019 -- 30-Dec-2019

Log Interval: 12/29/2019 8:00:00 AM -- 12/30/2019 2:30:00 AM

Index Types: Time

Index Scales: 5 cm / 3600 secs

Depth Source: Driller's Depth

Depth Sensor: 3rd Party Depth

Print Type: Final

Spud Date: 02-Dec-2019

Other Services:

Direction & Inclination (Surveys)

Directional Drilling

geoVISION Bit Resistivity

VISION Resistivity

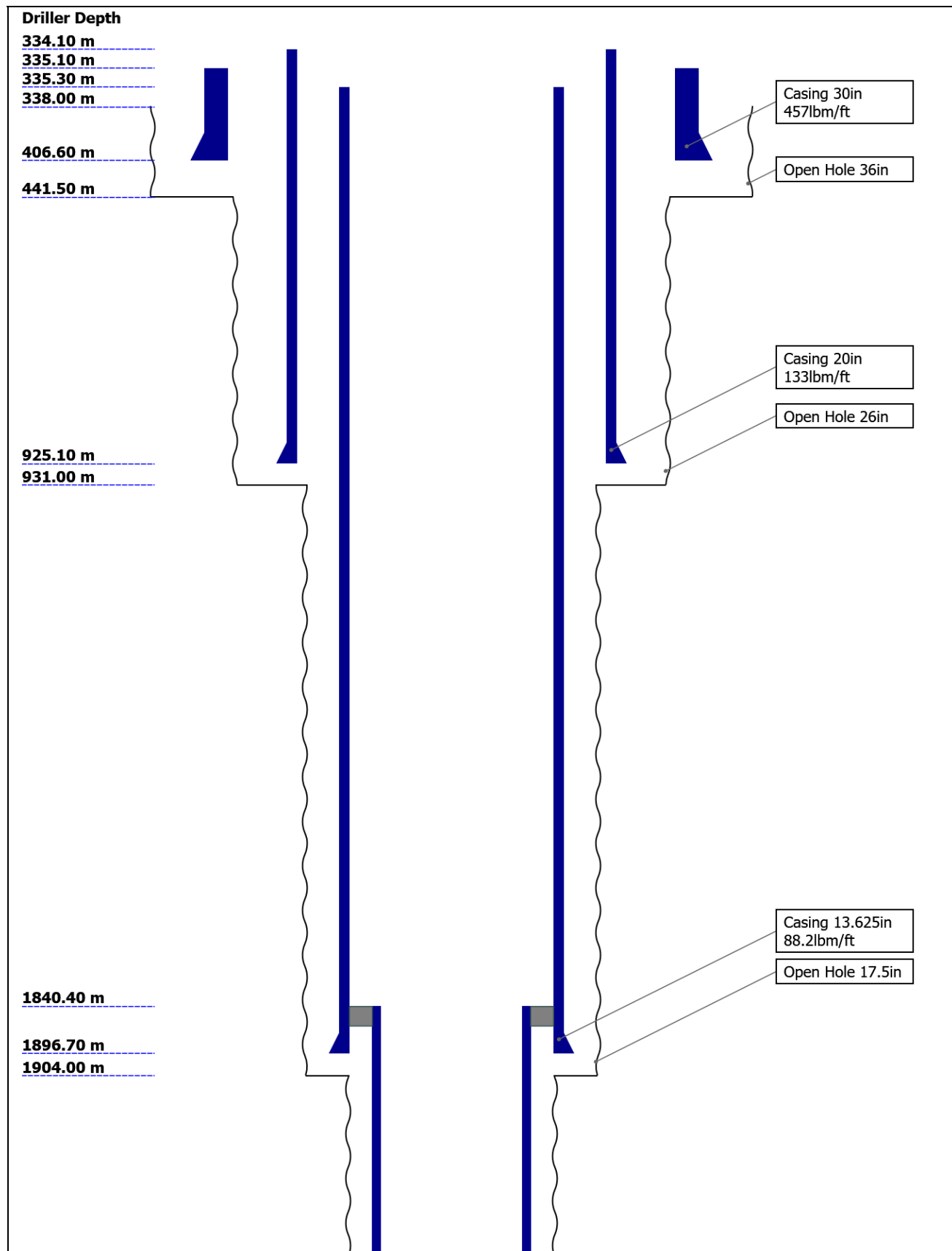
## Disclaimer

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## Well Sketch



2583.00 m

2584.00 m

2592.00 m

Casing 9.625in  
53.5lbm/ft

Open Hole 12.25in

Open Hole 8.5in

Borehole Size/Casing Record

|                       |        |       |        |        |      |  |
|-----------------------|--------|-------|--------|--------|------|--|
| Bit                   |        |       |        |        |      |  |
| Bit Size ( in )       | 36     | 26    | 17.5   | 12.25  | 8.5  |  |
| Top Driller ( m )     | 338    | 441.5 | 931    | 1904   | 2584 |  |
| Bottom Driller ( m )  | 441.5  | 931   | 1904   | 2584   | 2592 |  |
| Casing                |        |       |        |        |      |  |
| Size ( in )           | 30     | 20    | 13.625 | 9.625  |      |  |
| Weight ( lbm/ft )     | 457    | 133   | 88.2   | 53.5   |      |  |
| Inner Diameter ( in ) | 27.067 | 18.73 | 12.375 | 8.535  |      |  |
| Grade                 | X56    | N80   | P110   | P110   |      |  |
| Top Driller ( m )     | 335.1  | 334.1 | 335.3  | 1840.4 |      |  |
| Bottom Driller ( m )  | 406.6  | 925.1 | 1896.7 | 2583   |      |  |

Operational Run Summary

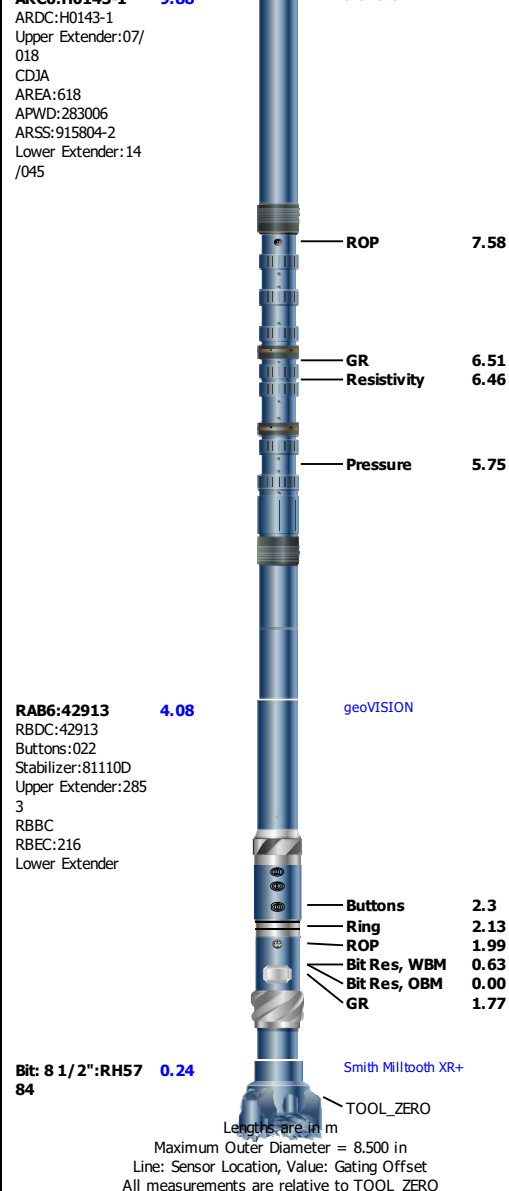
|                                  |                |  |  |  |  |  |
|----------------------------------|----------------|--|--|--|--|--|
| Parameter ( unit )               | Run 9          |  |  |  |  |  |
| Date Log Started                 | 29-Dec-2019    |  |  |  |  |  |
| Time Log Started                 | 05:27:58       |  |  |  |  |  |
| Date Log Finished                | 30-Dec-2019    |  |  |  |  |  |
| Time Log Finished                | 06:36:04       |  |  |  |  |  |
|                                  |                |  |  |  |  |  |
| Bit Size ( in )                  | 8.500          |  |  |  |  |  |
| Bit Start Depth ( m )            | 2584.00        |  |  |  |  |  |
| Bit Stop Depth ( m )             | 2592.00        |  |  |  |  |  |
| Top Log Interval ( m )           | 2584.70        |  |  |  |  |  |
| Bottom Log Interval ( m )        | 2591.22        |  |  |  |  |  |
|                                  |                |  |  |  |  |  |
| Max Hole Deviation ( deg )       | 1.10           |  |  |  |  |  |
| Azimuth of Max Deviation ( deg ) | 133.27         |  |  |  |  |  |
|                                  |                |  |  |  |  |  |
| Logging Unit Number              | A615           |  |  |  |  |  |
| Logging Unit Location            | Service Office |  |  |  |  |  |
| Recorded By                      | S.Kroevel      |  |  |  |  |  |
| Witnessed By                     | J.Eikenes      |  |  |  |  |  |
| Service Order Number             | 19SCA0085      |  |  |  |  |  |

Borehole Fluids

| Parameter( unit )                  | Run 9       |  |  |  |  |  |
|------------------------------------|-------------|--|--|--|--|--|
| Fluid Type                         | Water       |  |  |  |  |  |
| Fluid Name                         | Glydril     |  |  |  |  |  |
| Max Recorded Temperatures ( degC ) | 38          |  |  |  |  |  |
| Source of Sample                   | Active Tank |  |  |  |  |  |
| Salinity ( ppm )                   | 93984       |  |  |  |  |  |
| Density ( g/cm3 )                  | 1.15        |  |  |  |  |  |
| Funnel Viscosity ( s )             |             |  |  |  |  |  |
| Fluid Loss ( cm3 )                 |             |  |  |  |  |  |
| PH                                 |             |  |  |  |  |  |
| Source RMF                         |             |  |  |  |  |  |
| RMC                                | Pressed     |  |  |  |  |  |
| RM @ Meas Temp ( ohm.m@degC )      | 0.07 @ 18   |  |  |  |  |  |
| RMF @ Meas Temp ( ohm.m@degC )     | 0.06 @ 18   |  |  |  |  |  |
| RMC @ Meas Temp ( ohm.m@degC )     | 0.1 @ 20    |  |  |  |  |  |
| RM @ BHT ( ohm.m@degC )            | 0.05 @ 38   |  |  |  |  |  |
| RMF @ BHT ( ohm.m@degC )           | 0.04 @ 38   |  |  |  |  |  |
| RMC @ BHT ( ohm.m@degC )           | 0.07 @ 38   |  |  |  |  |  |
| Total Solid ( % )                  |             |  |  |  |  |  |
| High Gravity Solids ( % )          |             |  |  |  |  |  |

## Remarks and Equipment Summary

| Run 9: Toolstring                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                      |  |  | Run 9: Remarks |  |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|--|--|----------------|--|
| <div><div><div>Equip name</div><div>TELE675:E4388</div></div><div><div>Length</div><div>18.34</div></div><div><div>MP name</div><div>TeleScope</div></div><div><div>Offset</div><div></div></div></div> <div><div>MSSU675:30848</div><div>Upper Extender</div><div>MDC675:E4388</div><div>MMA:2507</div><div>MDI:3666</div><div>PMVC:236</div><div>PMEA:803</div><div>MTA:2162</div><div>MTK675:004</div><div>MSSD675:1005482-1</div><div>Lower Extender, Long</div></div> <div></div> <div><div>ARC6-H0143-1</div><div>9.88</div><div>arcVISION</div></div> | All depths referenced to driller's depth. See EOWR for depth tracking details.                                       |  |  |                |  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | APWD data from tool memory. All other data is Real Time.                                                             |  |  |                |  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Gamma Ray measurement is environmentally corrected for mud weight, bit size, collar thickness and Potassium content. |  |  |                |  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Run 9 POOH at 2592 m to perform coring.                                                                              |  |  |                |  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                      |  |  |                |  |



## Run 9

### Drilling Mechanics - APWD - Time Based

#### Pass Summary

| Run Name | Pass Objective | Direction | Start                  | Stop                   | Include Parallel Data |
|----------|----------------|-----------|------------------------|------------------------|-----------------------|
| Run 9    | TimeLogicalAcq | Down      | 29-Dec-2019 5:27:58 AM | 30-Dec-2019 6:36:04 AM | Yes                   |

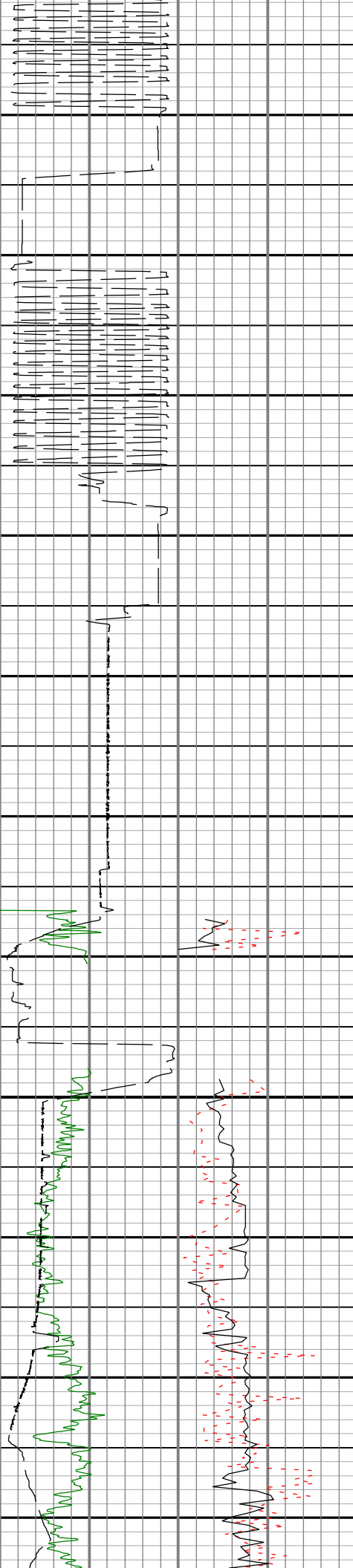
#### Log

Company:Equinor Well:31/5-7

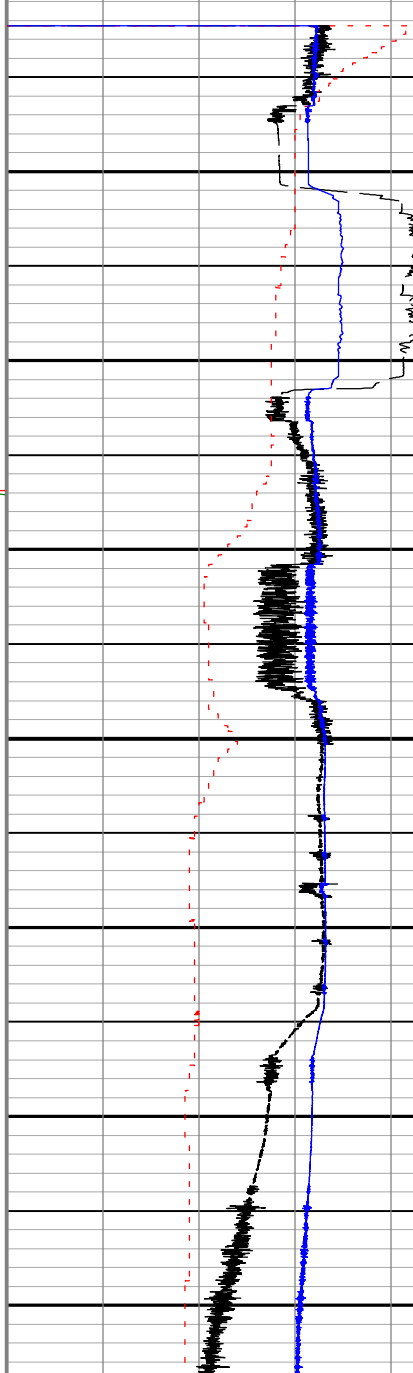
Run 9: TimeLogicalAcq:S053

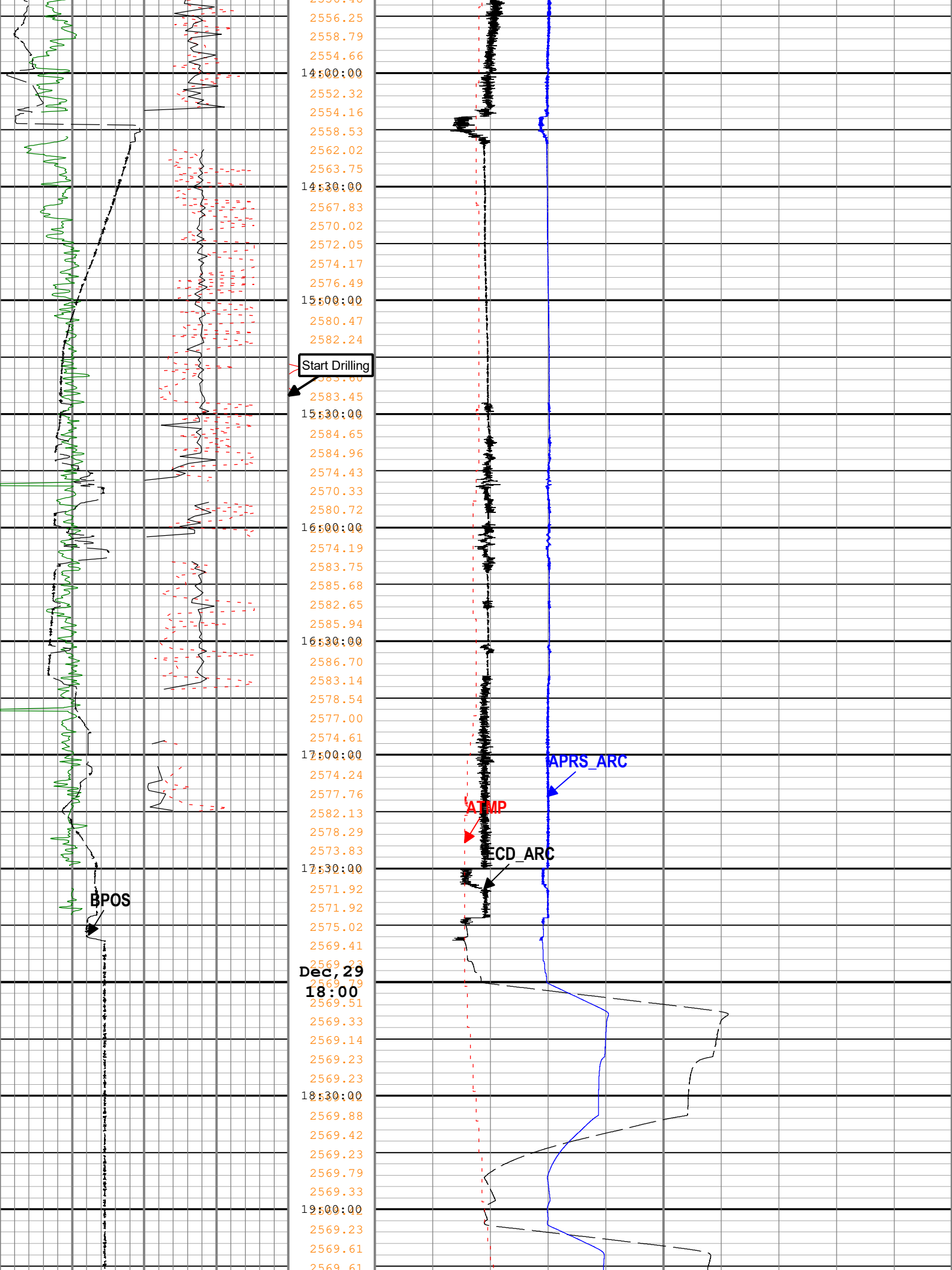
Description: Format: Log ( Drilling Mechanics Light ) Index Scale: 5 cm per 3600 s Index Type: Time Creation Date: 21-Jan-2020 10:02:01

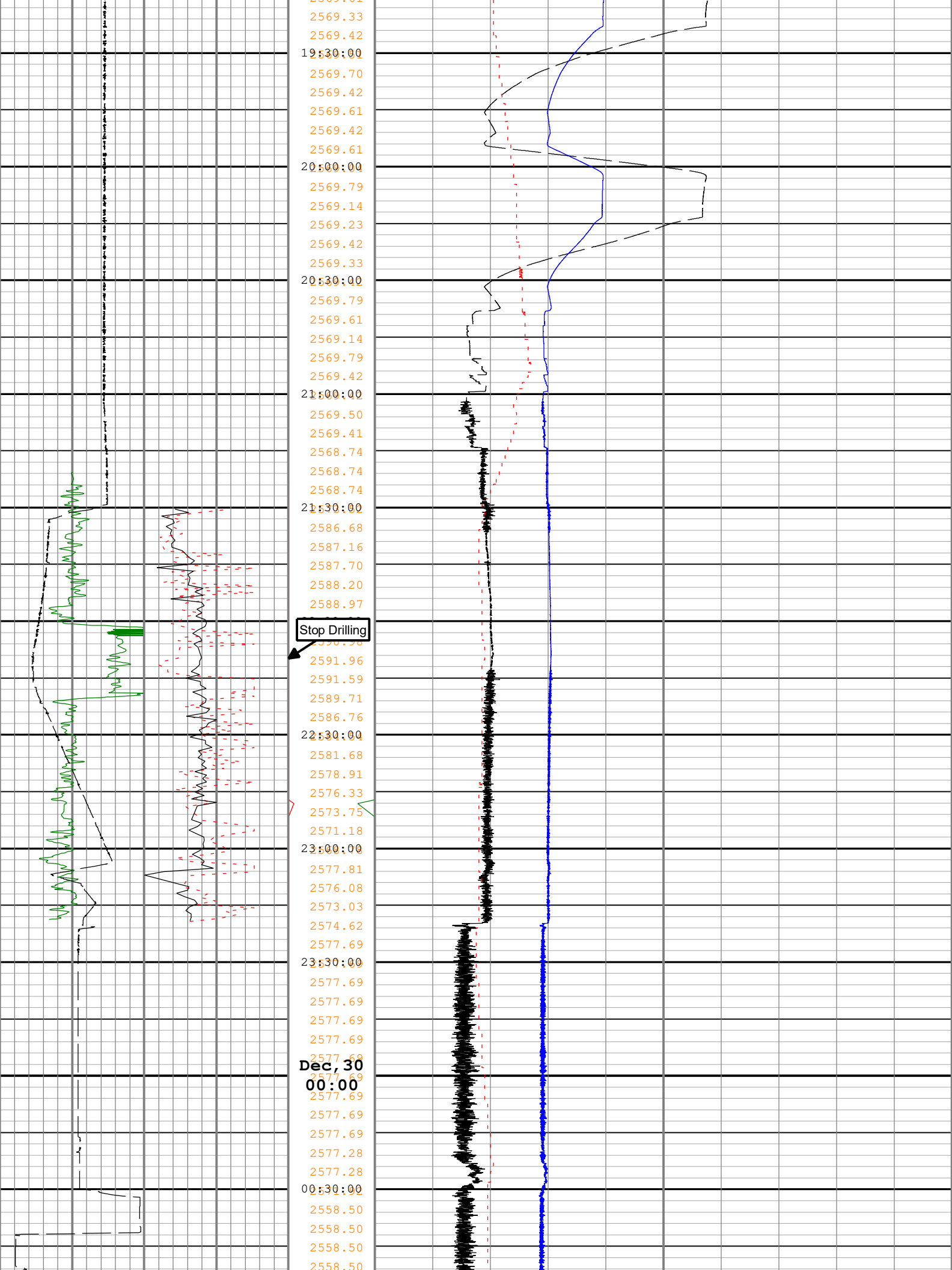
|                                              |                                                    | Depth Index<br>(DEPTH) RT<br>m              |                                                  |      |
|----------------------------------------------|----------------------------------------------------|---------------------------------------------|--------------------------------------------------|------|
| Gamma Ray<br>(GR_ARC_RT) ARC6<br>RT          | Stick Slip Indicator<br>(STICK_RT) TELE675<br>RT   | Shock Risk<br>(SHKRSK_R<br>T) TELE675<br>RT | Equivalent Circulating Density (ECD_ARC) ARC6 RM |      |
|                                              |                                                    |                                             | 1                                                | 2    |
| 0 gAPI 150                                   | 0 c/min 500                                        | 5 0                                         | g/cm3                                            |      |
| Height of block above<br>rig floor (BPOS) RT | Collar Rotational<br>Speed (CRPM_RT)<br>TELE675 RT | Shock Peak<br>(SHKPK_RT)<br>TELE675 RT      | Downhole Annulus Temperature (ATMP) ARC6 RM      |      |
|                                              |                                                    |                                             | 0                                                | 200  |
| 0 m 45                                       | 0 c/min 250                                        | 0 gn 250                                    | Downhole Annulus Pressure (APRS_ARC) ARC6 RM     |      |
|                                              |                                                    |                                             | bar                                              |      |
|                                              |                                                    |                                             | 0                                                | 1000 |

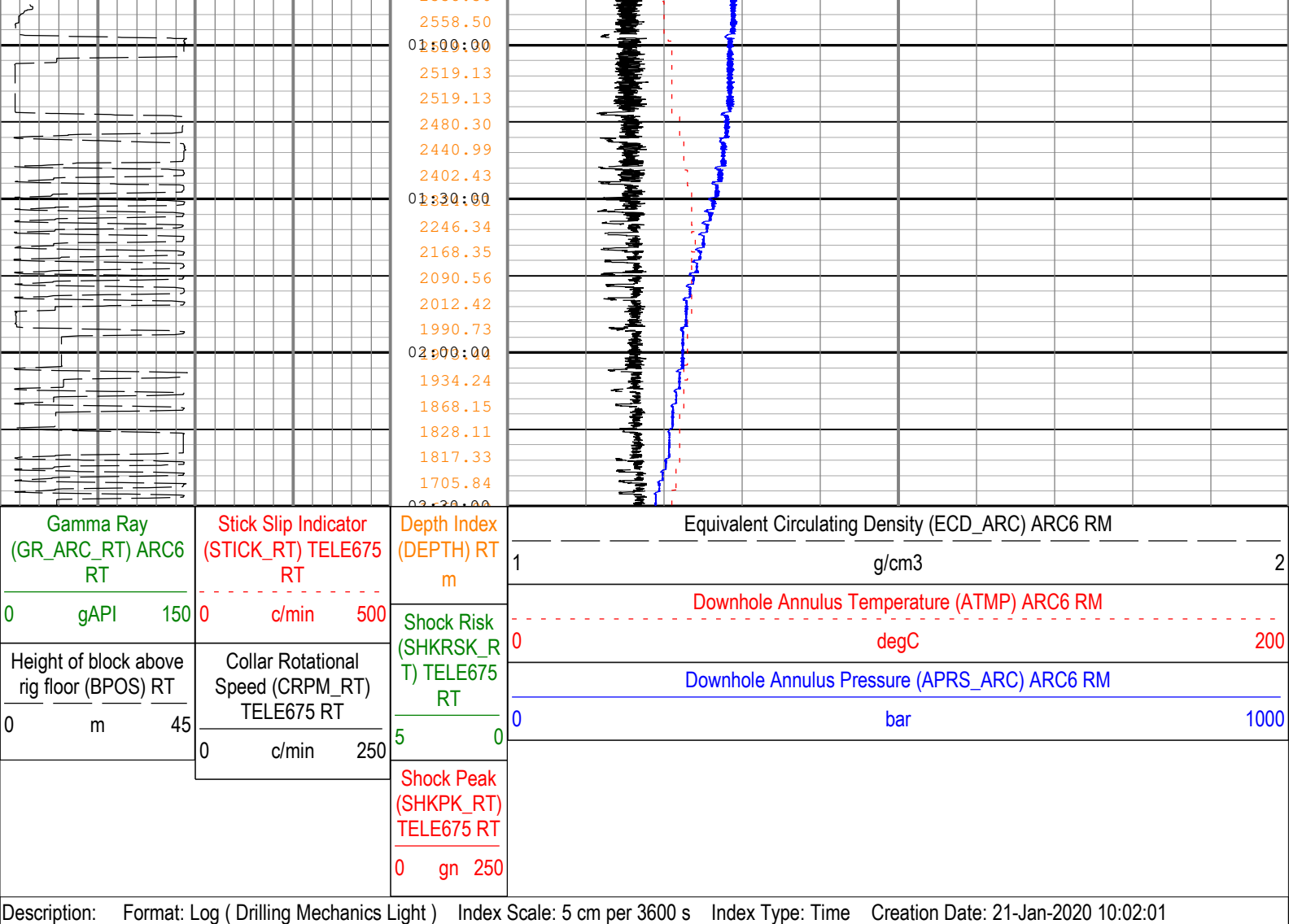


1489.63  
1566.68  
1643.63  
1719.88  
1797.42  
08:30:00  
1835.63  
1837.21  
1863.12  
1863.12  
1863.12  
09:00:00  
1894.18  
1944.39  
2021.65  
2134.63  
2214.20  
09:30:00  
2367.96  
2427.59  
2483.52  
2500.25  
2483.17  
10:00:00  
2483.05  
2483.05  
2491.71  
2495.63  
2502.00  
10:30:00  
2501.72  
2501.90  
2502.09  
2502.09  
2502.09  
11:00:00  
2502.27  
2501.83  
2496.28  
2495.90  
2510.63  
11:30:00  
2518.31  
2520.34  
2520.34  
2520.34  
2524.02  
Dec, 29  
12:00  
2549.74  
2553.16  
2553.21  
2553.61  
2553.52  
2553.62  
12:30:00  
2553.81  
2553.35  
2554.40  
2554.64  
2555.80  
13:00:00  
2559.34  
2561.13  
2559.12  
2557.62  
2555.22  
13:30:00  
2552.25  
2556.48









|           |               |
|-----------|---------------|
| Company:  | Equinor       |
| Well:     | 31/5-7        |
| Field:    | Eos           |
| Rig Name: | West Hercules |
| Country:  | Norway        |



